Internet & network security

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Data communication & networking
FIREWALL

NETWORK SECURITY TERMINOLOGIES



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MALWARE

It is a type of malicious program or software which can damage your system

= System Slow
(ii) file crash
(iii) ~~error~~ Style



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Types of malware



- Virus
- Worm
- Trojan horse
- Ransomware
- Spyware
- Keylogger
- Botnet
- Adware
- Rootkit

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Virus

It is type of malware which can attach itself

Viruses can be harmless or they can modify or delete data.

Its self replicant.



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worm

- It can run without host file
- It is self replicant
- It can damage remotely
- It slow down the system
-



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Trojan horse

- Trojan horse is any malware that misleads users of its true intent by disguising itself as a standard program
- A Trojan (or Trojan Horse) disguises itself as legitimate software with the purpose of tricking you into executing malicious software on your system ...
- Trojan horse does not replicate itself.
- The main objective of Trojan horse is to steal the information.

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spyware

Spyware is a kind of malware that secretly gathers information about a person or organization and relays this data to other parties without your knowledge or permission.



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Ransomware

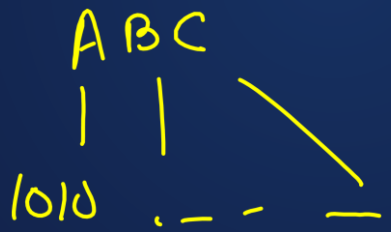
It is a type malware designed to deny a user or organization access to files on their computer. By encrypting these files and demanding a ransom amount for the decryption key.



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keylogger

Keylogger basically capturing the action of key stroke behavior .





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Rootkit

A rootkit is a type of malware that provides privileged, root-level (i.e., administrative) access to a computer

.

रूटकिट एक प्रकार का मैलवेयर है जो कंप्यूटर तक विशेषाधिकार प्राप्त, रूट-स्तरीय (यानी, प्रशासनिक) पहुंच प्रदान करता है

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Adware

Adware is the type of malware the term used for various pop-up advertisements that show up on your computer or mobile device.

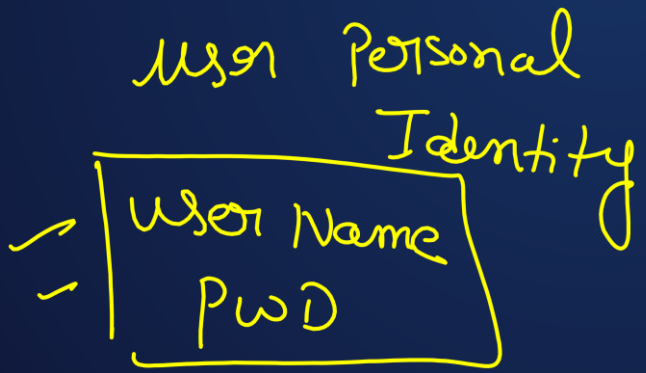
Unwanted Advertisment.



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Identity theft

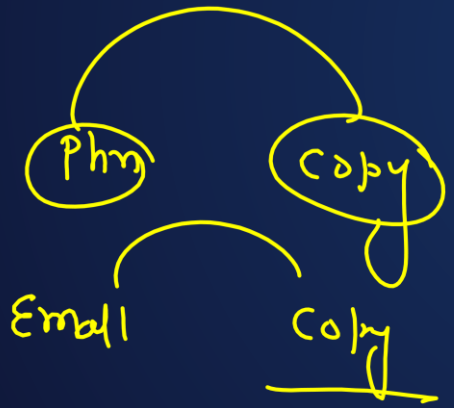
Identity theft refers to the unauthorized acquisition and use of someone else's personal information, typically for financial gain. The goal of identity theft is to assume the identity of the victim and carry out fraudulent activities.





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* Someunknow Person Prekend to be a Know Person.



Spoofing:

Spoofing,, is a technique used to deceive or trick individuals or computer systems by falsifying information.
It involves presenting false data or masquerading as someone or something else to gain unauthorized access or manipulate others.

Spoofing can be applied to various forms of communication, such as email, phone calls, websites, or IP addresses.



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Phishing

Phishing is a type of cyber attack in which an attacker attempts to deceive individuals or organizations into divulging sensitive information, such as login credentials, credit card details, or personal information,

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File Infector Viruses: These viruses attach themselves to executable files, such as .exe or .com files. When the infected file is executed, the virus activates and can infect other files on the system.

Macro Viruses: Macro viruses infect documents or files that contain macros, such as those in Microsoft Office documents (e.g., Word, Excel, PowerPoint). Once the infected document is opened, the virus can execute malicious macros and spread to other documents.

Application Program से damage

Boot Sector Viruses: These viruses infect the boot sector of storage devices (e.g., hard drives, USB drives) that are used to start up the computer. When the infected device is accessed during the boot process, the virus becomes active and can spread to other devices.

Hard Disk / floppy

Which Store :. Bootable file

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Firewall - which Prevent from Unauthorised attack.

Packet-Filtering Firewalls

- The most basic type of firewall, a packet-filtering firewall examines packets of data to determine whether they should be allowed through or blocked, based on rules like IP addresses, ports, and protocols.

- It checks the headers of incoming or outgoing packets and decides whether to allow or block traffic based on the configured rules.

- **Pros**: Simple, fast, and low resource consumption.
- **Cons**: Cannot inspect the content of packets, making it vulnerable to certain types of attacks (e.g., IP spoofing).



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Stateful Inspection Firewalls

•Stateful firewalls are more advanced than packet-filtering firewalls. They track the state of active connections and make decisions based on the context of the traffic (e.g., whether a packet is part of an established connection or a new request).

•: It keeps track of the state of active connections (e.g., TCP connection states) and allows or blocks traffic based on this context.

•**Pros**: More secure than packet-filtering firewalls, as they understand the state of the connection.
•**Cons**: More resource-intensive and complex compared to packet filtering.

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Proxy Firewalls (Application Layer Firewalls)



- Proxy firewalls act as an intermediary between the client and the server. They inspect traffic at the application layer (Layer 7 of the OSI model) and can perform deep packet inspection (DPI).

- **How it works**: When a client sends a request, the proxy firewall intercepts and forwards it to the server, and vice versa. It can filter traffic based on more specific application-layer criteria, such as specific websites or applications.

- Can block specific content or applications (e.g., blocking specific websites or file types) and provide detailed logging.
- **Cons**: Performance can be impacted due to the overhead of inspecting traffic at the application layer.

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Next-Generation Firewalls (NGFW)

- NGFWs combine traditional firewall capabilities with additional security features like intrusion prevention systems (IPS), deep packet inspection (DPI), and application awareness.

- They inspect traffic not just at the network level but also at the application level, allowing for more granular control over traffic based on user identity, application type, and more.

- Provides comprehensive security, including malware detection, deep inspection, and prevention of advanced threats.

- Can be more complex to configure and requires more resources than traditional firewalls.

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Circuit-Level Gateways

•These firewalls work at the session layer (Layer 5) and monitor the establishment and termination of connections. They do not inspect the content of packets but instead focus on the state of the connection.

•When a connection is established, the circuit-level gateway verifies the integrity and trust of the connection, allowing data transfer to proceed once the connection is validated.

•**Pros**: Fast and less resource-intensive than application-layer firewalls.
•**Cons**: Limited in terms of filtering content and more vulnerable to certain types of attacks.

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Host-Based Firewalls

- **Overview**: These firewalls are software-based and installed directly on individual devices (e.g., servers, workstations, or laptops). They monitor and control traffic to and from the specific host.

- **How it works**: The firewall monitors the traffic to and from the host and blocks or allows packets based on local rules.

- **Pros**: Provides protection for individual devices, preventing unauthorized access at the device level.

- **Cons**: They don’t provide protection for network-wide traffic and require configuration on each individual device.

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Network-Based Firewalls

•: Network-based firewalls are typically hardware devices placed between a trusted internal network and untrusted external networks (e.g., the internet).

•: These firewalls examine all incoming and outgoing traffic between the internal network and the external network, applying rules to decide whether to allow or block specific traffic.
•**Pros**: Provide centralized security for the entire network and can handle high traffic loads.
•**Cons**: Not as granular as host-based firewalls and may require more resources.

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Cloud Firewalls

- Also known as "Firewall-as-a-Service" (FWaaS), cloud firewalls are hosted in the cloud rather than on-premises, and they provide network security for cloud-based environments.

- Cloud firewalls work similarly to traditional firewalls but are hosted and managed by a cloud provider. They provide protection for cloud infrastructure, applications, and services.
- **Pros**: Scalable, easy to manage, and can protect against threats in dynamic cloud environments.
- **Cons**: Relies on internet connectivity and can introduce latency.

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Distributed Firewalls

- A distributed firewall is typically used in virtualized environments and is implemented as part of a virtual machine (VM) or software-defined network (SDN).

- These firewalls apply security rules to virtual networks or individual VMs, providing a more granular and scalable approach in environments with dynamic workloads.
- **Pros**: Flexible, scalable, and well-suited to modern, cloud-based or virtualized infrastructures.
- **Cons**: Complexity increases with scale and requires careful management of virtual network configurations.

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Hybrid Firewalls

- Hybrid firewalls combine features of multiple firewall types, such as stateful inspection, application filtering, and intrusion prevention systems (IPS), to offer a comprehensive security solution.
- By integrating multiple layers of security, hybrid firewalls can offer robust defense against a wide range of attack vectors.
- **Pros**: Combines the strengths of different firewall types, providing robust protection.
- **Cons**: Complex to configure and maintain, and may require more resources.

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Which of the following is true about IPv6?

- A) It uses a 32-bit address space.
- B) It supports header compression to reduce packet size.
- C) It does not support multicast addressing.
- D) It has a fixed length of 64 bits for all addresses.

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What is the purpose of ICMP in the network layer?

- A) To establish connections between hosts.
- B) To provide error messages and operational information.
- C) To manage congestion in the network.
- D) To encapsulate transport layer segments into packets.

Internet Control message Protocol

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Which protocol is primarily used for path MTU discovery?

- A) TCP — Connection oriented
- B) ICMP
- C) UDP — Connection less
- D) ARP — IP - MAC

maximum transmission unit

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What is the main difference between TCP and UDP?

- A) TCP is connection-oriented, while UDP is connectionless.
- B) TCP is faster than UDP.
- C) UDP guarantees delivery, while TCP does not.
- D) TCP uses less bandwidth than UDP.

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Which of the following mechanisms is used by TCP to ensure reliable delivery of data?

- A) Handshaking
- B) Checksums
- C) Sequence numbers and acknowledgments
- D) All of the above

In which scenario would you prefer using UDP over TCP?

- A) File transfer
- B) Streaming video
- C) Email transmission
- D) Web page loading

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Which of the following protocols is used for the transport of voice over IP (VoIP) communications?

- A) RTP (Real-time Transport Protocol)
- B) SMTP (Simple Mail Transfer Protocol)
- C) FTP (File Transfer Protocol)
- D) SNMP (Simple Network Management Protocol)

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In the context of data communication, what is "jitter"?

- A) The delay in packet delivery.
- B) The variation in packet arrival times.
- C) The loss of packets during transmission.
- D) The total bandwidth used by the connection.

What is the primary purpose of a checksum in data transmission?

- A) To encrypt the data for security.
- B) To provide redundancy for error correction.
- C) To detect errors in transmitted data.
- D) To compress the data before transmission.

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Which layer of the OSI model is responsible for session management in data communication?

- A) Transport Layer
- B) Application Layer
- C) Session Layer
- D) Network Layer

In a wireless communication system, what does "fade margin" refer to?

- A) The difference between received signal strength and noise level.
- B) The maximum allowable loss in a transmission line.
- C) The difference between the signal strength and the threshold required for acceptable communication.
- D) The minimum bandwidth required for a specific application.

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Which technique is used to improve bandwidth efficiency in digital communication systems?

- A) Time Division Multiplexing (TDM)
- B) Frequency Shift Keying (FSK)
- C) Amplitude Modulation (AM)
- D) Spread Spectrum

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What is the role of the ARP (Address Resolution Protocol) in data communication?

- A) To translate IP addresses into MAC addresses.
- B) To route packets across different networks.
- C) To manage the flow of data between devices.
- D) To encrypt data for secure transmission.

ARP IP – MAC

RARP = MAC – IP

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1. Which of the following best describes a Next-Generation Firewall (NGFW)?

A) A firewall that only performs packet filtering at the network layer.

B) A firewall that integrates multiple layers of security, such as IDS/IPS, deep packet inspection, and application awareness.

C) A firewall that filters traffic based on IP addresses and ports.

D) A firewall that operates only at the transport layer and performs connection tracking.

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What is the main advantage of a proxy firewall over a packet-filtering firewall?

A) It performs deep packet inspection to examine the contents of the traffic.
B) It operates faster by inspecting only the headers of packets.
C) It filters traffic at the network layer only, improving performance.
D) It requires fewer resources and is easier to configure.

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Which of the following statements is true about a Stateful Inspection Firewall?

A) It only checks the headers of the packets without considering the state of the connection.
B) It maintains a table of active connections and inspects traffic based on the connection's state.
C) It can only filter based on IP addresses and ports.
D) It operates solely at the application layer and inspects the content of traffic.

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What does a Web Application Firewall (WAF) primarily protect against?

A) Denial-of-Service (DoS) attacks
B) Internal network attacks
C) Application-layer vulnerabilities such as SQL injection and cross-site scripting (XSS)
D) Network-layer attacks like packet sniffing

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Which of the following is a disadvantage of using a packet-filtering firewall?

A) It can filter traffic based on application layer protocols.
B) It cannot inspect the contents of the packets, making it vulnerable to certain attacks.
C) It consumes more system resources than stateful inspection firewalls.
D) It requires deep packet inspection, which can be computationally expensive.

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What is the primary function of an Intrusion Prevention System (IPS) integrated into a Next-Generation Firewall (NGFW)?

A) To log incoming traffic for future analysis.
B) To prevent unauthorized access by blocking packets based on IP and port.
C) To actively analyze traffic and block potential threats in real-time.
D) To act as a reverse proxy for external web servers.

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A Distributed Denial of Service (DDoS) attack targets a web application. Which type of firewall is best suited to protect against this attack?

A) Proxy Firewall
B) Network Firewall
C) Web Application Firewall (WAF)
D) Stateful Inspection Firewall

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What type of address is used by a switch to forward data to the correct destination in a local area network (LAN)?

IP address

Port number

URL

MAC address

None of the above

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Which network topology provides high fault tolerance and scalability but requires more cabling than other topologies?

Mesh Topology
Star Topology
Ring Topology
Bus Topology
Tree Topology

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In a traditional network setup, what is the primary function of a hub as a network device?

To intelligently route traffic between different networks

To filter and forward data based on MAC addresses

To connect multiple devices and broadcast data to all connected devices

To provide firewall protection for network traffic

To allocate IP addresses to devices within a network

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When discussing network devices, which device operates at Layer 2 of the OSI model and forwards data packets based on MAC addresses?

Network Switches

Router

Bridge

Repeater

None of the above

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In the context of network communication, what is the primary purpose of a port in TCP/IP networking?

To identify the hardware device connected to the network

To specify the encryption protocol used for communication

To differentiate between multiple applications or services running on the same IP address

To assign IP addresses to devices in the network
E
To provide physical access to the network for devices

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In IPv4 addressing, given the IP address 192.168.16.45/28, what is the subnet mask and the number of usable hosts in this subnet?

255.255.255.240, 30 hosts
255.255.255.248, 14 hosts
255.255.255.240, 14 hosts
255.255.255.224, 30 hosts
255.255.255.248, 6 hosts

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In the context of networking, what is the primary role of ARP (Address Resolution Protocol), and how does it function when a host in a LAN wants to communicate with another device on the same network?

Resolves MAC addresses into IP addresses

Broadcasts an ARP request to the network to map an IP address to a MAC address

Directly routes packets between different subnets

Encrypts data for secure communication

Detects and prevents network loops within the LAN

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What is the primary function of a DNS server, and how does it resolve domain names in a typical query?

Converts MAC addresses into IP addresses
Maps human-readable domain names to IP addresses
Encrypts web traffic for secure communication
Authenticates users accessing network resources
E
Balances load across multiple servers in a network

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What is the primary purpose of FTP (File Transfer Protocol) in networking?

Encrypting network traffic

Transferring files between a client and a server

Allocating IP addresses to network devices

Managing email communication

Securing communication between two hosts

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What is the purpose of a firewall in a computer network?

To connect different networks together
To provide wireless connectivity
To increase the speed of data transmission
To filter and block unauthorized access to and from the network
None of the above

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